常 用 积 分 公 式

（一）含有 $ax+b$ 的积分（$a\neq 0$）

1. $\int \frac{\mathrm{d}x}{ax+b}=\frac{1}{a}\ln|ax+b|+C$

2. $\int (ax+b)^{\mu}\mathrm{d}x=\frac{1}{a(\mu+1)}(ax+b)^{\mu+1}+C$（$\mu\neq -1$）

3. $\int \frac{x}{ax+b}\mathrm{d}x=\frac{1}{a^2}(ax+b-b\ln|ax+b|)+C$

4. $\int \frac{x^2}{ax+b}\mathrm{d}x=\frac{1}{a^3}\left[\frac{1}{2}(ax+b)^2-2b(ax+b)+b^2\ln|ax+b|\right]+C$

5. $\int \frac{\mathrm{d}x}{x(ax+b)}=-\frac{1}{b}\ln\left|\frac{ax+b}{x}\right|+C$

6. $\int \frac{\mathrm{d}x}{x^2(ax+b)}=-\frac{1}{bx}+\frac{a}{b^2}\ln\left|\frac{ax+b}{x}\right|+C$

7. $\int \frac{x}{(ax+b)^2}\mathrm{d}x=\frac{1}{a^2}(\ln|ax+b|+\frac{b}{ax+b})+C$

8. $\int \frac{x^2}{(ax+b)^2}\mathrm{d}x=\frac{1}{a^3}(ax+b-2b\ln|ax+b|-\frac{b^2}{ax+b})+C$

9. $\int \frac{\mathrm{d}x}{x(ax+b)^2}=\frac{1}{b(ax+b)}-\frac{1}{b^2}\ln\left|\frac{ax+b}{x}\right|+C$

（二）含有 $\sqrt{ax+b}$ 的积分

10. $\int \sqrt{ax+b}\mathrm{d}x=\frac{2}{3a}\sqrt{(ax+b)^3}+C$

11. $\int x\sqrt{ax+b}\mathrm{d}x=\frac{2}{15a^2}(3ax-2b)\sqrt{(ax+b)^3}+C$

12. $\int x^2\sqrt{ax+b}\mathrm{d}x=\frac{2}{105a^3}(15a^2x^2-12abx+8b^2)\sqrt{(ax+b)^3}+C$

13. $\int \frac{x}{\sqrt{ax+b}}\mathrm{d}x=\frac{2}{3a^2}(ax-2b)\sqrt{ax+b}+C$

14. $\int \frac{x^2}{\sqrt{ax+b}}\mathrm{d}x = \frac{2}{15a^3}(3a^2x^2 - 4abx + 8b^2)\sqrt{ax+b} + C$

15. $\int \frac{\mathrm{d}x}{x\sqrt{ax+b}} = \begin{cases} \frac{1}{\sqrt{b}}\ln\left|\frac{\sqrt{ax+b}-\sqrt{b}}{\sqrt{ax+b}+\sqrt{b}}\right| + C & (b>0) \\ \frac{2}{\sqrt{-b}}\arctan\sqrt{\frac{ax+b}{-b}} + C & (b<0) \end{cases}$

16. $\int \frac{\mathrm{d}x}{x^2\sqrt{ax+b}} = -\frac{\sqrt{ax+b}}{bx} - \frac{a}{2b}\int \frac{\mathrm{d}x}{x\sqrt{ax+b}}$

17. $\int \frac{\sqrt{ax+b}}{x}\mathrm{d}x = 2\sqrt{ax+b} + b\int \frac{\mathrm{d}x}{x\sqrt{ax+b}}$

18. $\int \frac{\sqrt{ax+b}}{x^2}\mathrm{d}x = -\frac{\sqrt{ax+b}}{x} + \frac{a}{2}\int \frac{\mathrm{d}x}{x\sqrt{ax+b}}$

（三）含有 $x^2 \pm a^2$ 的积分

19. $\int \frac{\mathrm{d}x}{x^2+a^2} = \frac{1}{a}\arctan\frac{x}{a} + C$

20. $\int \frac{\mathrm{d}x}{(x^2+a^2)^n} = \frac{x}{2(n-1)a^2(x^2+a^2)^{n-1}} + \frac{2n-3}{2(n-1)a^2}\int \frac{\mathrm{d}x}{(x^2+a^2)^{n-1}}$

21. $\int \frac{\mathrm{d}x}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C$

（四）含有 $ax^2+b(a>0)$ 的积分

22. $\int \frac{\mathrm{d}x}{ax^2+b} = \begin{cases} \frac{1}{\sqrt{ab}}\arctan\sqrt{\frac{a}{b}}\,x + C & (b>0) \\ \frac{1}{2\sqrt{-ab}}\ln\left|\frac{\sqrt{a}x-\sqrt{-b}}{\sqrt{a}x+\sqrt{-b}}\right| + C & (b<0) \end{cases}$

23. $\int \frac{x}{ax^2+b}\mathrm{d}x = \frac{1}{2a}\ln\left|ax^2+b\right| + C$

24. $\int \frac{x^2}{ax^2+b}\mathrm{d}x = \frac{x}{a} - \frac{b}{a}\int \frac{\mathrm{d}x}{ax^2+b}$

25. $\int \frac{\mathrm{d}x}{x(ax^2+b)} = \frac{1}{2b}\ln\frac{x^2}{|ax^2+b|}+C$

26. $\int \frac{\mathrm{d}x}{x^2(ax^2+b)} = -\frac{1}{bx}-\frac{a}{b}\int\frac{\mathrm{d}x}{ax^2+b}$

27. $\int \frac{\mathrm{d}x}{x^3(ax^2+b)} = \frac{a}{2b^2}\ln\frac{|ax^2+b|}{x^2}-\frac{1}{2bx^2}+C$

28. $\int \frac{\mathrm{d}x}{(ax^2+b)^2} = \frac{x}{2b(ax^2+b)}+\frac{1}{2b}\int\frac{\mathrm{d}x}{ax^2+b}$

（五）含有 $ax^2+bx+c\ (a>0)$ 的积分

29. $$\int \frac{\mathrm{d}x}{ax^2+bx+c} = \begin{cases} \frac{2}{\sqrt{4ac-b^2}}\arctan\frac{2ax+b}{\sqrt{4ac-b^2}}+C & (b^2<4ac) \\ \frac{1}{\sqrt{b^2-4ac}}\ln\left|\frac{2ax+b-\sqrt{b^2-4ac}}{2ax+b+\sqrt{b^2-4ac}}\right|+C & (b^2>4ac) \end{cases}$$

30. $\int \frac{x}{ax^2+bx+c}\mathrm{d}x = \frac{1}{2a}\ln|ax^2+bx+c|-\frac{b}{2a}\int\frac{\mathrm{d}x}{ax^2+bx+c}$

（六）含有 $\sqrt{x^2+a^2}\ (a>0)$ 的积分

31. $\int \frac{\mathrm{d}x}{\sqrt{x^2+a^2}} = \operatorname{arsh}\frac{x}{a}+C_1 = \ln(x+\sqrt{x^2+a^2})+C$

32. $\int \frac{\mathrm{d}x}{\sqrt{(x^2+a^2)^3}} = \frac{x}{a^2\sqrt{x^2+a^2}}+C$

33. $\int \frac{x}{\sqrt{x^2+a^2}}\mathrm{d}x = \sqrt{x^2+a^2}+C$

34. $\int \frac{x}{\sqrt{(x^2+a^2)^3}}\mathrm{d}x = -\frac{1}{\sqrt{x^2+a^2}}+C$

35. $\int \frac{x^2}{\sqrt{x^2+a^2}}\mathrm{d}x = \frac{x}{2}\sqrt{x^2+a^2}-\frac{a^2}{2}\ln(x+\sqrt{x^2+a^2})+C$

36. $\int \frac{x}{\sqrt{(x^2+a^2)^3}}\mathrm{d}x = -\frac{x}{\sqrt{x^2+a^2}} + \ln(x+\sqrt{x^2+a^2}) + C$

37. $\int \frac{\mathrm{d}x}{x\sqrt{x^2+a^2}} = \frac{1}{a}\ln\frac{\sqrt{x^2+a^2}-a}{|x|} + C$

38. $\int \frac{\mathrm{d}x}{x^2\sqrt{x^2+a^2}} = -\frac{\sqrt{x^2+a^2}}{a^2x} + C$

39. $\int \sqrt{x^2+a^2}\mathrm{d}x = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\ln(x+\sqrt{x^2+a^2}) + C$

40. $\int \sqrt{(x^2+a^2)^3}\mathrm{d}x = \frac{x}{8}(2x^2+5a^2)\sqrt{x^2+a^2} + \frac{3}{8}a^4\ln(x+\sqrt{x^2+a^2}) + C$

41. $\int x\sqrt{x^2+a^2}\mathrm{d}x = \frac{1}{3}\sqrt{(x^2+a^2)^3} + C$

42. $\int x^2\sqrt{x^2+a^2}\mathrm{d}x = \frac{x}{8}(2x^2+a^2)\sqrt{x^2+a^2} - \frac{a^4}{8}\ln(x+\sqrt{x^2+a^2}) + C$

43. $\int \frac{\sqrt{x^2+a^2}}{x}\mathrm{d}x = \sqrt{x^2+a^2} + a\ln\frac{\sqrt{x^2+a^2}-a}{|x|} + C$

44. $\int \frac{\sqrt{x^2+a^2}}{x^2}\mathrm{d}x = -\frac{\sqrt{x^2+a^2}}{x} + \ln(x+\sqrt{x^2+a^2}) + C$

（七）含有 $\sqrt{x^2-a^2}$ $(a>0)$ 的积分

45. $\int \frac{\mathrm{d}x}{\sqrt{x^2-a^2}} = \frac{x}{|x|}\mathrm{arch}\frac{|x|}{a} + C_1 = \ln\left|x+\sqrt{x^2-a^2}\right| + C$

46. $\int \frac{\mathrm{d}x}{\sqrt{(x^2-a^2)^3}} = -\frac{x}{a^2\sqrt{x^2-a^2}} + C$

47. $\int \frac{x}{\sqrt{x^2-a^2}}\mathrm{d}x = \sqrt{x^2-a^2} + C$

48. $\int \frac{x}{\sqrt{(x^2-a^2)^3}}\mathrm{d}x = -\frac{1}{\sqrt{x^2-a^2}} + C$

49. $\int \frac{x^2}{\sqrt{x^2-a^2}}\mathrm{d}x = \frac{x}{2}\sqrt{x^2-a^2} + \frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right| + C$

50. $\int \frac{x^2}{\sqrt{(x^2-a^2)^3}}\mathrm{d}x = -\frac{x}{\sqrt{x^2-a^2}} + \ln\left|x+\sqrt{x^2-a^2}\right| + C$

51. $\int \frac{\mathrm{d}x}{x\sqrt{x^2-a^2}} = \frac{1}{a}\arccos\frac{a}{|x|} + C$

52. $\int \frac{\mathrm{d}x}{x^2\sqrt{x^2-a^2}} = \frac{\sqrt{x^2-a^2}}{a^2x} + C$

53. $\int \sqrt{x^2-a^2}\,\mathrm{d}x = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln\left|x+\sqrt{x^2-a^2}\right| + C$

54. $\int \sqrt{(x^2-a^2)^3}\,\mathrm{d}x = \frac{x}{8}(2x^2-5a^2)\sqrt{x^2-a^2} + \frac{3}{8}a^4\ln\left|x+\sqrt{x^2-a^2}\right| + C$

55. $\int x\sqrt{x^2-a^2}\,\mathrm{d}x = \frac{1}{3}\sqrt{(x^2-a^2)^3} + C$

56. $\int x^2\sqrt{x^2-a^2}\,\mathrm{d}x = \frac{x}{8}(2x^2-a^2)\sqrt{x^2-a^2} - \frac{a^4}{8}\ln\left|x+\sqrt{x^2-a^2}\right| + C$

57. $\int \frac{\sqrt{x^2-a^2}}{x}\mathrm{d}x = \sqrt{x^2-a^2} - a\arccos\frac{a}{|x|} + C$

58. $\int \frac{\sqrt{x^2-a^2}}{x^2}\mathrm{d}x = -\frac{\sqrt{x^2-a^2}}{x} + \ln\left|x+\sqrt{x^2-a^2}\right| + C$

（八）含有 $\sqrt{a^2-x^2}$ $(a>0)$ 的积分

59. $\int \frac{\mathrm{d}x}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a} + C$

60. $\int \frac{\mathrm{d}x}{\sqrt{(a^2-x^2)^3}} = \frac{x}{a^2\sqrt{a^2-x^2}} + C$

61. $\int \frac{x}{\sqrt{a^2-x^2}}\mathrm{d}x = -\sqrt{a^2-x^2} + C$

62. $\int \frac{x}{\sqrt{(a^2-x^2)^3}}\mathrm{d}x = \frac{1}{\sqrt{a^2-x^2}} + C$

63. $\int \frac{x^2}{\sqrt{a^2-x^2}}\mathrm{d}x = -\frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\arcsin\frac{x}{a} + C$

64. $\int \frac{x^2}{\sqrt{(a^2-x^2)^3}}\mathrm{d}x = \frac{x}{\sqrt{a^2-x^2}} - \arcsin\frac{x}{a} + C$

65. $\int \frac{\mathrm{d}x}{x\sqrt{a^2-x^2}} = \frac{1}{a}\ln\frac{a-\sqrt{a^2-x^2}}{|x|} + C$

66. $\int \frac{\mathrm{d}x}{x^2\sqrt{a^2-x^2}} = -\frac{\sqrt{a^2-x^2}}{a^2x} + C$

67. $\int \sqrt{a^2-x^2}\mathrm{d}x = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\arcsin\frac{x}{a} + C$

68. $\int \sqrt{(a^2-x^2)^3}\mathrm{d}x = \frac{x}{8}(5a^2-2x^2)\sqrt{a^2-x^2} + \frac{3}{8}a^4\arcsin\frac{x}{a} + C$

69. $\int x\sqrt{a^2-x^2}\mathrm{d}x = -\frac{1}{3}\sqrt{(a^2-x^2)^3} + C$

70. $\int x^2\sqrt{a^2-x^2}\mathrm{d}x = \frac{x}{8}(2x^2-a^2)\sqrt{a^2-x^2} + \frac{a^4}{8}\arcsin\frac{x}{a} + C$

71. $\int \frac{\sqrt{a^2-x^2}}{x}\mathrm{d}x = \sqrt{a^2-x^2} + a\ln\frac{a-\sqrt{a^2-x^2}}{|x|} + C$

72. $\int \frac{\sqrt{a^2-x^2}}{x^2}\mathrm{d}x = -\frac{\sqrt{a^2-x^2}}{x} - \arcsin\frac{x}{a} + C$

（九）含有 $\sqrt{\pm ax^2+bx+c}$ $(a>0)$ 的积分

73. $\int \frac{\mathrm{d}x}{\sqrt{ax^2+bx+c}} = \frac{1}{\sqrt{a}}\ln\left|2ax+b+2\sqrt{a}\sqrt{ax^2+bx+c}\right| + C$

74. $\int \sqrt{ax^2+bx+c}\mathrm{d}x = \frac{2ax+b}{4a}\sqrt{ax^2+bx+c}$

$$+\frac{4ac-b^2}{8\sqrt{a^3}}\ln\left|2ax+b+2\sqrt{a}\sqrt{ax^2+bx+c}\right|+C$$

75. $\int\frac{x}{\sqrt{ax^2+bx+c}}\mathrm{d}x=\frac{1}{a}\sqrt{ax^2+bx+c}$

$$-\frac{b}{2\sqrt{a^3}}\ln\left|2ax+b+2\sqrt{a}\sqrt{ax^2+bx+c}\right|+C$$

76. $\int\frac{\mathrm{d}x}{\sqrt{c+bx-ax^2}}=-\frac{1}{\sqrt{a}}\arcsin\frac{2ax-b}{\sqrt{b^2+4ac}}+C$

77. $\int\sqrt{c+bx-ax^2}\mathrm{d}x=\frac{2ax-b}{4a}\sqrt{c+bx-ax^2}+\frac{b^2+4ac}{8\sqrt{a^3}}\arcsin\frac{2ax-b}{\sqrt{b^2+4ac}}+C$

78. $\int\frac{x}{\sqrt{c+bx-ax^2}}\mathrm{d}x=-\frac{1}{a}\sqrt{c+bx-ax^2}+\frac{b}{2\sqrt{a^3}}\arcsin\frac{2ax-b}{\sqrt{b^2+4ac}}+C$

（十）含有 $\sqrt{\pm\frac{x-a}{x-b}}$ 或 $\sqrt{(x-a)(b-x)}$ 的积分

79. $\int\sqrt{\frac{x-a}{x-b}}\mathrm{d}x=(x-b)\sqrt{\frac{x-a}{x-b}}+(b-a)\ln(\sqrt{|x-a|}+\sqrt{|x-b|})+C$

80. $\int\sqrt{\frac{x-a}{b-x}}\mathrm{d}x=(x-b)\sqrt{\frac{x-a}{b-x}}+(b-a)\arcsin\sqrt{\frac{x-a}{b-x}}+C$

81. $\int\frac{\mathrm{d}x}{\sqrt{(x-a)(b-x)}}=2\arcsin\sqrt{\frac{x-a}{b-x}}+C\qquad(a<b)$

82. $\int\sqrt{(x-a)(b-x)}\mathrm{d}x=\frac{2x-a-b}{4}\sqrt{(x-a)(b-x)}+\frac{(b-a)^2}{4}\arcsin\sqrt{\frac{x-a}{b-x}}+C$

$(a<b)$

（十一）含有三角函数的积分

83. $\int\sin x\mathrm{d}x=-\cos x+C$

84. $\int\cos x\mathrm{d}x=\sin x+C$

85. $\int \tan x\mathrm{d}x = -\ln|\cos x| + C$

86. $\int \cot x\mathrm{d}x = \ln|\sin x| + C$

87. $\int \sec x\mathrm{d}x = \ln\left|\tan(\frac{\pi}{4}+\frac{x}{2})\right| + C = \ln|\sec x + \tan x| + C$

88. $\int \csc x\mathrm{d}x = \ln\left|\tan\frac{x}{2}\right| + C = \ln|\csc x - \cot x| + C$

89. $\int \sec^2 x\mathrm{d}x = \tan x + C$

90. $\int \csc^2 x\mathrm{d}x = -\cot x + C$

91. $\int \sec x\tan x\mathrm{d}x = \sec x + C$

92. $\int \csc x\cot x\mathrm{d}x = -\csc x + C$

93. $\int \sin^2 x\mathrm{d}x = \frac{x}{2} - \frac{1}{4}\sin 2x + C$

94. $\int \cos^2 x\mathrm{d}x = \frac{x}{2} + \frac{1}{4}\sin 2x + C$

95. $\int \sin^n x\mathrm{d}x = -\frac{1}{n}\sin^{n-1} x\cos x + \frac{n-1}{n}\int \sin^{n-2} x\mathrm{d}x$

96. $\int \cos^n x\mathrm{d}x = \frac{1}{n}\cos^{n-1} x\sin x + \frac{n-1}{n}\int \cos^{n-2} x\mathrm{d}x$

97. $\int \frac{\mathrm{d}x}{\sin^n x} = -\frac{1}{n-1}\cdot\frac{\cos x}{\sin^{n-1} x} + \frac{n-2}{n-1}\int \frac{\mathrm{d}x}{\sin^{n-2} x}$

98. $\int \frac{\mathrm{d}x}{\cos^n x} = \frac{1}{n-1}\cdot\frac{\sin x}{\cos^{n-1} x} + \frac{n-2}{n-1}\int \frac{\mathrm{d}x}{\cos^{n-2} x}$

99. $\int \cos^m x\sin^n x\mathrm{d}x = \frac{1}{m+n}\cos^{m-1} x\sin^{n+1} x + \frac{m-1}{m+n}\int \cos^{m-2} x\sin^n x\mathrm{d}x$

$= -\frac{1}{m+n}\cos^{m+1} x\sin^{n-1} x + \frac{n-1}{m+n}\int \cos^m x\sin^{n-2} x\mathrm{d}x$

100. $\int \sin ax\cos bx\mathrm{d}x = -\frac{1}{2(a+b)}\cos(a+b)x - \frac{1}{2(a-b)}\cos(a-b)x + C$

101. $\int \sin ax\sin bx\mathrm{d}x = -\frac{1}{2(a+b)}\sin(a+b)x + \frac{1}{2(a-b)}\sin(a-b)x + C$

102. $\int \cos ax \cos bx \mathrm{d}x = \frac{1}{2(a+b)}\sin(a+b)x + \frac{1}{2(a-b)}\sin(a-b)x + C$

103. $\int \frac{\mathrm{d}x}{a+b\sin x} = \frac{2}{\sqrt{a^2-b^2}}\arctan\frac{a\tan\frac{x}{2}+b}{\sqrt{a^2-b^2}} + C \quad (a^2 > b^2)$

104. $\int \frac{\mathrm{d}x}{a+b\sin x} = \frac{1}{\sqrt{b^2-a^2}}\ln\left|\frac{a\tan\frac{x}{2}+b-\sqrt{b^2-a^2}}{a\tan\frac{x}{2}+b+\sqrt{b^2-a^2}}\right| + C \quad (a^2 < b^2)$

105. $\int \frac{\mathrm{d}x}{a+b\cos x} = \frac{2}{a+b}\sqrt{\frac{a+b}{a-b}}\arctan(\sqrt{\frac{a-b}{a+b}}\tan\frac{x}{2}) + C \quad (a^2 > b^2)$

106. $\int \frac{\mathrm{d}x}{a+b\cos x} = \frac{1}{a+b}\sqrt{\frac{a+b}{b-a}}\ln\left|\frac{\tan\frac{x}{2}+\sqrt{\frac{a+b}{b-a}}}{\tan\frac{x}{2}-\sqrt{\frac{a+b}{b-a}}}\right| + C \quad (a^2 < b^2)$

107. $\int \frac{\mathrm{d}x}{a^2\cos^2 x + b^2\sin^2 x} = \frac{1}{ab}\arctan(\frac{b}{a}\tan x) + C$

108. $\int \frac{\mathrm{d}x}{a^2\cos^2 x - b^2\sin^2 x} = \frac{1}{2ab}\ln\left|\frac{b\tan x + a}{b\tan x - a}\right| + C$

109. $\int x\sin ax \mathrm{d}x = \frac{1}{a^2}\sin ax - \frac{1}{a}x\cos ax + C$

110. $\int x^2\sin ax \mathrm{d}x = -\frac{1}{a}x^2\cos ax + \frac{2}{a^2}x\sin ax + \frac{2}{a^3}\cos ax + C$

111. $\int x\cos ax \mathrm{d}x = \frac{1}{a^2}\cos ax + \frac{1}{a}x\sin ax + C$

112. $\int x^2\cos ax \mathrm{d}x = \frac{1}{a}x^2\sin ax + \frac{2}{a^2}x\cos ax - \frac{2}{a^3}\sin ax + C$

（十二）含有反三角函数的积分（其中 $a > 0$ ）

113. $\int \arcsin\frac{x}{a}\mathrm{d}x = x\arcsin\frac{x}{a} + \sqrt{a^2-x^2} + C$

114. $\int x\arcsin\frac{x}{a}\mathrm{d}x = (\frac{x^2}{2} - \frac{a^2}{4})\arcsin\frac{x}{a} + \frac{x}{4}\sqrt{a^2-x^2} + C$

115. $\int x^2 \arcsin\frac{x}{a}\mathrm{d}x = \frac{x^3}{3}\arcsin\frac{x}{a} + \frac{1}{9}(x^2+2a^2)\sqrt{a^2-x^2} + C$

116. $\int \arccos\frac{x}{a}\mathrm{d}x = x\arccos\frac{x}{a} - \sqrt{a^2-x^2} + C$

117. $\int x\arccos\frac{x}{a}\mathrm{d}x = (\frac{x^2}{2} - \frac{a^2}{4})\arccos\frac{x}{a} - \frac{x}{4}\sqrt{a^2-x^2} + C$

118. $\int x^2 \arccos\frac{x}{a}\mathrm{d}x = \frac{x^3}{3}\arccos\frac{x}{a} - \frac{1}{9}(x^2+2a^2)\sqrt{a^2-x^2} + C$

119. $\int \arctan\frac{x}{a}\mathrm{d}x = x\arctan\frac{x}{a} - \frac{a}{2}\ln(a^2+x^2) + C$

120. $\int x\arctan\frac{x}{a}\mathrm{d}x = \frac{1}{2}(a^2+x^2)\arctan\frac{x}{a} - \frac{a}{2}x + C$

121. $\int x^2 \arctan\frac{x}{a}\mathrm{d}x = \frac{x^3}{3}\arctan\frac{x}{a} - \frac{a}{6}x^2 + \frac{a^3}{6}\ln(a^2+x^2) + C$

（十三）含有指数函数的积分

122. $\int a^x \mathrm{d}x = \frac{1}{\ln a}a^x + C$

123. $\int \mathrm{e}^{ax}\mathrm{d}x = \frac{1}{a}\mathrm{e}^{ax} + C$

124. $\int x\mathrm{e}^{ax}\mathrm{d}x = \frac{1}{a^2}(ax-1)\mathrm{e}^{ax} + C$

125. $\int x^n\mathrm{e}^{ax}\mathrm{d}x = \frac{1}{a}x^n\mathrm{e}^{ax} - \frac{n}{a}\int x^{n-1}\mathrm{e}^{ax}\mathrm{d}x$

126. $\int xa^x\mathrm{d}x = \frac{x}{\ln a}a^x - \frac{1}{(\ln a)^2}a^x + C$

127. $\int x^n a^x\mathrm{d}x = \frac{1}{\ln a}x^n a^x - \frac{n}{\ln a}\int x^{n-1}a^x\mathrm{d}x$

128. $\int \mathrm{e}^{ax}\sin bx\mathrm{d}x = \frac{1}{a^2+b^2}\mathrm{e}^{ax}(a\sin bx - b\cos bx) + C$

129. $\int \mathrm{e}^{ax}\cos bx\mathrm{d}x = \frac{1}{a^2+b^2}\mathrm{e}^{ax}(b\sin bx + a\cos bx) + C$

130. $\int \mathrm{e}^{ax}\sin^n bx\mathrm{d}x = \frac{1}{a^2+b^2n^2}\mathrm{e}^{ax}\sin^{n-1}bx(a\sin bx - nb\cos bx)$

$$+\frac{n(n-1)b^2}{a^2+b^2n^2}\int e^{ax}\sin^{n-2}bx\mathrm{d}x$$

131. $\int e^{ax}\cos^n bx\mathrm{d}x=\frac{1}{a^2+b^2n^2}e^{ax}\cos^{n-1}bx(a\cos bx+nb\sin bx)$

$$+\frac{n(n-1)b^2}{a^2+b^2n^2}\int e^{ax}\cos^{n-2}bx\mathrm{d}x$$

（十四）含有对数函数的积分

132. $\int \ln x\mathrm{d}x=x\ln x-x+C$

133. $\int\frac{\mathrm{d}x}{x\ln x}=\ln|\ln x|+C$

134. $\int x^n\ln x\mathrm{d}x=\frac{1}{n+1}x^{n+1}(\ln x-\frac{1}{n+1})+C$

135. $\int(\ln x)^n\mathrm{d}x=x(\ln x)^n-n\int(\ln x)^{n-1}\mathrm{d}x$

136. $\int x^m(\ln x)^n\mathrm{d}x=\frac{1}{m+1}x^{m+1}(\ln x)^n-\frac{n}{m+1}\int x^m(\ln x)^{n-1}\mathrm{d}x$

（十五）含有双曲函数的积分

137. $\int \mathrm{sh}x\mathrm{d}x=\mathrm{ch}x+C$

138. $\int \mathrm{ch}x\mathrm{d}x=\mathrm{sh}x+C$

139. $\int \mathrm{th}x\mathrm{d}x=\ln\mathrm{ch}x+C$

140. $\int \mathrm{sh}^2x\mathrm{d}x=-\frac{x}{2}+\frac{1}{4}\mathrm{sh}2x+C$

141. $\int \mathrm{ch}^2x\mathrm{d}x=\frac{x}{2}+\frac{1}{4}\mathrm{sh}2x+C$

（十六）定积分

142. $\int_{-\pi}^{\pi}\cos nx\mathrm{d}x=\int_{-\pi}^{\pi}\sin nx\mathrm{d}x=0$

143. $\int_{-\pi}^{\pi}\cos mx\sin nx\mathrm{d}x=0$

144. $\int_{-\pi}^{\pi}\cos mx\cos nx\mathrm{d}x=\begin{cases}0, & m\neq n\\ \pi, & m=n\end{cases}$

145. $\int_{-\pi}^{\pi}\sin mx\sin nx\mathrm{d}x=\begin{cases}0, & m\neq n\\ \pi, & m=n\end{cases}$

146. $\int_{0}^{\pi}\sin mx\sin nx\mathrm{d}x=\int_{0}^{\pi}\cos mx\cos nx\mathrm{d}x=\begin{cases}0, & m\neq n\\ \dfrac{\pi}{2}, & m=n\end{cases}$

147. $I_n=\int_0^{\frac{\pi}{2}}\sin^n x\mathrm{d}x=\int_0^{\frac{\pi}{2}}\cos^n x\mathrm{d}x$

$I_n=\dfrac{n-1}{n}I_{n-2}$

$I_n=\dfrac{n-1}{n}\cdot\dfrac{n-3}{n-2}\cdot\cdots\cdot\dfrac{4}{5}\cdot\dfrac{2}{3}$ （n 为大于 1 的正奇数），$I_1=1$

$I_n=\dfrac{n-1}{n}\cdot\dfrac{n-3}{n-2}\cdot\cdots\cdot\dfrac{3}{4}\cdot\dfrac{1}{2}\cdot\dfrac{\pi}{2}$ （n 为正偶数），$I_0=\dfrac{\pi}{2}$